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Practice article

Optimized neural-network-assisted model-based motion control of hydraulic manipulators under unmodeled uncertainties

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HIGHLIGHTS

- NN-assisted backstepping control for hydraulic manipulators uses K-means++ optimized RBFNNs to handle nonlinear dynamics.
- Integrates weight modification and uses desired trajectory signals to suppress sensor noise and enhance control robustness.
- Comparative experiments on a hydraulic manipulator verify superior tracking performance relative to advanced existing methods.

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ABSTRACT

Multi-degree-of-freedom (multi-DoF) hydraulic manipulators are widely used in heavy-duty tasks due to their high power density. However, their inherent nonlinearities and uncertain dynamics present significant challenges to precise control. Traditional model-based control methods, while effective in addressing these nonlinearities, rely heavily on accurate dynamic models, which complicates controller design and increases computational demands. To address these limitations, this paper proposes a neural-network-assisted adaptive robust control approach that reduces the dependence on precise modeling by compensating for unmodeled dynamics. Radial basis function neural networks (RBFNNs) are employed to approximate the uncertain dynamics, with the network structure optimized using the K-means++ algorithm to enhance approximation accuracy and computational efficiency. Additionally, desired signals are used instead of measured signals to mitigate sensitivity to measurement noise. The proposed method is rigorously analyzed to guarantee the stability and asymptotic tracking performance of the closed-loop system. Experimental results on a hydraulic manipulator validate the effectiveness of the approach, demonstrating substantial improvements in control performance compared with conventional model-based strategies.

1. Introduction

Multi-degree-of-freedom (multi-DoF) hydraulic manipulators are widely used in industries such as manufacturing, aerospace, and heavy machinery [1–4]. Their high power density, excellent load-carrying capacity, and flexible dynamic response characteristics make them particularly suitable for tasks requiring large forces [5,6]. However, with the continuous improvement of accuracy requirements, the challenge of high-precision control continues to emerge [7–9]. Firstly, hydraulic systems exhibit significant nonlinearities due to fluid dynamics and valve characteristics, which complicate accurate modeling of pressure and flow dynamics in hydraulic manipulators [10,11]. Additionally, inherent dynamics of hydraulic actuators, such as fluid compressibility and non-constant flow characteristics, introduce further complexity compared to motor-driven systems.

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Secondly, parameter uncertainties, such as variations in friction coefficient due to temperature and pressure changes, increase system complexity. If not properly addressed, these uncertainties can degrade performance and cause system instability [12,13]. Traditional control methods, such as proportional-integral-derivative (PID) control and fuzzy logic control, have been widely used in hydraulic systems [14–16]. However, these methods often neglect actuator dynamics and complex nonlinearities, resulting in large steady-state errors and instability, especially in high-precision applications.

To address these limitations, model-based control methods have emerged. For instance, friction torque at the joints has been extensively studied. Han et al. [17] applied a trigonometric nonlinear function model and higher-order velocity terms to characterize viscous friction variations, while Lee et al. [18] further employed a LuGre model to more accurately capture friction characteristics during start–stop motions and extremely low-speed conditions. However, these studies primarily focused on system identification rather than control design, as increased model complexity introduces additional challenges for controller design. Based on these studies, Xu et al. [19] improved control performance in hydraulic manipulators by incorporating a parameterized Stribeck model. J. Yao et al. [20,21] introduced a hyperbolic tangent function to represent the LuGre model, addressing non-differentiability issues and enhancing performance in hydraulic actuators. Xia et al. [13] proposed an improved LuGre friction model that ensures continuity and differentiability, which significantly enhanced control performance in low-speed operations of multi-DoF hydraulic manipulators.

Although deterministic model-based control has demonstrated success in hydraulic manipulator applications, where model compensation is only achieved through inverse dynamics and does not update parameters, its performance heavily depends on the accuracy of modeling. To mitigate the limitations of deterministic model-based control, adaptive model control methods have garnered increasing attention. B. Yao et al. [11,22] proposed an adaptive robust motion control theory to address the challenges of controlling single-rod hydraulic actuators while managing parameter uncertainties and unmodeled nonlinearities in hydraulic systems. Based on this, the desired compensation adaptive robust controller was developed to mitigate the adverse effects of measurement noise on control performance [23,24], J. Yao et al. [25,26] conducted extensive research on the parameter uncertainty in hydraulic systems and developed robust integral of the sign of the error (RISE) controller to effectively suppress parameter uncertainty and reduce tracking errors of hydraulic systems. Despite promising results on single-actuator platforms, these methods still face challenges in multi-DoF manipulators due to joint coupling and more complex dynamics. Zhou et al. [27] developed an observer-based adaptive robust controller for a multi-DoF hydraulic manipulator subject to both parametric uncertainties and the lack of accurate velocity measurements.

In recent years, researchers have begun exploring the integration of neural networks (NNs) into adaptive robust control frameworks to compensate for the limitations of traditional model-compensation-based methods [28]. Radial Basis Function Neural Networks (RBFNNs), known for their powerful function approximation capabilities, have been widely used for modeling and controlling nonlinear systems [29,30]. While foundational methods like Fourier series and classical Bernstein polynomials provided early analytical approaches for function approximation in control, the advent of more sophisticated function approximation techniques (FATs)—such as the Mastroianni operators and the (p,q)-analogue Bernstein operators—has demonstrated significant value. These advanced operators, owing to their the inherent mathematical simplicity, offer efficient solutions for observer-based or regressor-free adaptive controllers. Nevertheless, for systems with complex state-dependent nonlinearities and unmodeled dynamics, such as hydraulic manipulators, local approximators like RBFNNs remain a mainstream and efficient choice due to their flexible structure and strong local approximation capability. For example, J. Yao et al. introduced NN algorithms into the RISE controller to improve the accuracy of model-based feedforward terms [31,32], and Kim et al. applied NN algorithms to permanent magnet synchronous motors, achieving promising experimental results [33]. Jin et al. utilized RBFNNs to optimize fault diagnosis in a dual-disk hollow shaft rotor system [34], and other scholars tried to combine sliding mode control with RBFNNs to handle dynamic uncertainty in spacecraft [35,36]. This is because RBFNNs, due to their single-hidden-layer architecture and low parameter count, offer a good trade-off between approximation accuracy and computational efficiency, making them well-suited for small-sample learning and real-time signal processing tasks [37]. Moreover, their weight updates rely solely on current input data, enabling fast online learning without storing full data histories, which is advantageous for real-time control. These characteristics make RBFNNs particularly suitable for resource-constrained platforms, such as embedded or edge systems [38]. However, the performance of RBFNNs heavily relies on the selection of centers and width parameters [39]. Traditional methods for choosing these parameters often depend on trial-and-error or empirical approaches, which are inefficient and prone to suboptimal solutions in high-dimensional spaces [40]. By leveraging the K-means++ algorithm, these parameters can be effectively optimized based on prior data, resulting in better input space representation, efficient computation of width parameters, and improved overall performance, making RBFNNs more efficient in practical applications [41].

To further enhance control performance through the utilization of NNs, this study proposes an optimized neural-network-assisted adaptive robust control approach that takes advantage of NNs to improve model-based control methods for hydraulic manipulators. The contributions of this article can be summarized as follows.

(1) A novel NN-assisted backstepping control framework is developed for high-order nonlinear dynamics of a multi-DoF hydraulic manipulator. The compensation for system uncertainties is realized using RBFNNs, with their center distributions optimized via the K-means++ algorithm to enhance approximation capability and online learning efficiency.
(2) The robustness of the system is substantially enhanced by integrating a modification term into the RBFNN weight adaptation process, and the desired trajectory signals are employed instead of directly using measured sensor signals in the controller design, effectively suppressing sensor noise and external disturbances and thus strengthening the closed-loop robustness of the proposed controller.
(3) Comparative experiments on a hydraulic manipulator platform demonstrate that the proposed control strategy achieves superior tracking performance relative to existing methods.

Notation: To facilitate expression and for the convenience of matching the dimensions of matrix vectors in the formula, we define two operators $\odot$ and $\oslash$, with the following forms:

$$\alpha \odot \beta = [\alpha_1, \alpha_2, \dots, \alpha_n]^{\mathrm{T}} \odot [\beta_1, \beta_2, \dots, \beta_n]^{\mathrm{T}} = [\alpha_1\beta_1, \alpha_2\beta_2, \dots, \alpha_n\beta_n]^{\mathrm{T}},$$

$$\alpha \oslash \beta = [\alpha_1, \alpha_2, \dots, \alpha_n]^{\mathrm{T}} \oslash [\beta_1, \beta_2, \dots, \beta_n]^{\mathrm{T}} = [\alpha_1/\beta_1, \alpha_2/\beta_2, \dots, \alpha_n/\beta_n]^{\mathrm{T}}.$$

2. Dynamic model and problem formulation

2.1. Dynamics of an n-DoF rigid manipulator

To establish the universality of the proposed dynamic model, this section examines an n-DoF hydraulic manipulator. As depicted in Fig. 1,

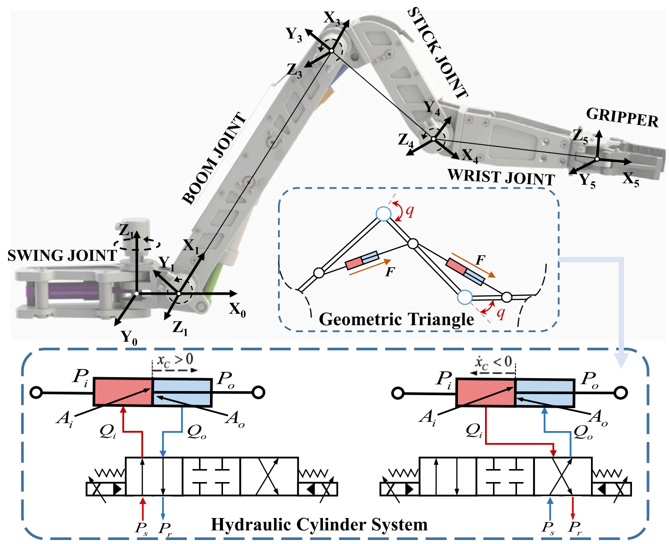

Fig. 1. Schematic diagram of the considered multi-DoF hydraulic manipulator.

each joint of the manipulator is precisely controlled by hydraulic actuators, and is meticulously regulated through proportional valves. The dynamics of the manipulator can be described as follows:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau - \tau_f(\dot{q}) + \tau_\varepsilon(t), \tag{1}$$

where $M(q) \in \mathbb{R}^{n\times n}$ denotes the inertia matrix, which is a symmetric positive-definite matrix, $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ represents the Coriolis and centrifugal torque, while $G(q) \in \mathbb{R}^n$ denotes the vector of gravity torque; $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ represent the angular position, angular velocity, and angular acceleration of the manipulator joints, respectively, $\tau \in \mathbb{R}^n$ is the output torque of the hydraulic cylinder acting on the joint, which will be elaborated upon later in relation to the structural characteristics of the hydraulic manipulator, and $\tau_\varepsilon(t) \in \mathbb{R}^n$ represents the modeling error, including modeling uncertainty and external interference.

The friction torque vector $\tau_f(\dot{q}) \in \mathbb{R}^n$ can be expressed as

$$\tau_f(\dot{q}) = f_{\dot{q}} \odot \dot{q} + f_\mu \odot S_f(\dot{q}), \tag{2}$$

where $f_{\dot{q}} \in \mathbb{R}^n$ and $f_\mu \in \mathbb{R}^n$ represent the viscous and the Coulomb friction coefficients, respectively, $S_f(\bullet) \in \mathbb{R}^n$ is a discontinuous sign function. Unlike traditional hydraulic actuators, the joint angle of a hydraulic manipulator does not follow a linear path relative to the movement of the piston within the hydraulic cylinder [13]. Therefore, we use the Jacobian matrix $J_h(q)$ to define the relationship between them [19,42], and the output torque τ can be further written as

$$\tau = J_h(q)F = J_h(q)\left(P_i \odot A_i - P_o \odot A_o\right), \tag{3}$$

$$J_h(q) = diag\left[J_{h1}(q_1), J_{h2}(q_2), \ldots, J_{hn}(q_n)\right], \tag{4}$$

$$J_{hi}(q) = \frac{L_{1i}L_{2i}\sin(q_i)}{\sqrt{L_{1i}^2 + L_{2i}^2 - 2L_{1i}L_{2i}\cos(q_i)}}, \tag{5}$$

where $F \in \mathbb{R}^n$ is the thrust of hydraulic cylinder, $P_i \in \mathbb{R}^n$ and $P_o \in \mathbb{R}^n$ represent the pressures within each chamber of the cylinder, $A_i \in \mathbb{R}^n$ and $A_o \in \mathbb{R}^n$ are the areas of each cylinder, L_{1i}, L_{2i}, and q_i respectively represent the triangle edge lengths and angle of the i-th joint. The relationship between the cylinder displacement x_C, speed $\dot{x}_C$ and joint motion can be written as:

$$\dot{x}_C = J_h(q)\dot{q}. \tag{6}$$

Property 1. The inertia matrix $M(q)$ is a positive definite symmetric matrix, and satisfies

$$\underline{\delta}\|v\|^2 \le v^{\mathrm{T}} M(q) v \le \bar{\delta}\|v\|^2,\ \forall v \in \mathbb{R}^n, \tag{7}$$

with $\underline{\delta}$ and $\bar{\delta}$ being the minimum and maximum eigenvalues of $M(q)$, respectively, where v is an arbitrary vector.

Property 2. The derivative of $M(q)$ and the matrix $C(q,\dot{q})$ satisfy

$$v^{\mathrm{T}}[\dot{M}(q) - 2C(q,\dot{q})]v = 0,\ \forall v \in \mathbb{R}^n. \tag{8}$$

2.2. Dynamics of hydraulic system

The n-DoF manipulator is driven by hydraulic actuators. To ensure the general applicability of the conclusions, the pressure-flow dynamics of the hydraulic actuator can be expressed as

$$\begin{aligned} V_i \odot \dot{P}_i &= \beta_e\left[-A_i \odot \dot{x}_C + Q_i - \sigma\left(P_i - P_o\right) + d_{Q_i}(t)\right], \\ V_o \odot \dot{P}_o &= \beta_e\left[A_o \odot \dot{x}_C - Q_o - \sigma\left(P_o - P_i\right) + d_{Q_o}(t)\right], \end{aligned} \tag{9}$$

where $V_i \in \mathbb{R}^n$ and $V_o \in \mathbb{R}^n$ denote the head- and rod-end chamber volumes of each cylinder, $\beta_e \in \mathbb{R}$ is the effective bulk modulus, $Q_i \in \mathbb{R}^n$ and $Q_o \in \mathbb{R}^n$ denote the supplied flow rates of each cylinder, $\sigma \in \mathbb{R}$ is the oil leakage coefficient, and $d_{Q_i}(t) \in \mathbb{R}^n$ and $d_{Q_o}(t) \in \mathbb{R}^n$ represent the uncertain nonlinearities in flow rates.

To ensure a rapid system response, the hydraulic power unit is configured as a constant pressure source. Consequently, this analysis concentrates solely on the dynamics of the valve-controlled hydraulic cylinder. While some studies [22] have considered the internal dynamics of the servo valve, such modeling typically requires an additional sensor to detect the spool position. This adds system complexity with only limited benefits in motion tracking accuracy. Therefore, many existing works, including [13], omit the servo valve dynamics altogether. This work adopts a high-response servo valve, which allows for the reasonable assumption of a linear relationship between the control input voltage $u_v \in \mathbb{R}^n$ and the spool displacement, thus simplifying the system model.

$$\begin{aligned} Q_i &= k_{u_i} \odot g_i(P_i, u_v) \odot u_v, \\ Q_o &= k_{u_o} \odot g_o(P_o, u_v) \odot u_v, \end{aligned} \tag{10}$$

where $k_{u_i} \in \mathbb{R}^n$ and $k_{u_o} \in \mathbb{R}^n$ are the constant flow gain coefficients matrices of the forward and return loops respectively, $g_i(P_i, u_v) \in \mathbb{R}^n$ and $g_o(P_o, u_v) \in \mathbb{R}^n$ can be expressed as

$$\begin{aligned} g_i(P_i, u_v) &= \begin{cases} (P_s - P_i)^{\frac{1}{2}}, u_{v_n} \ge 0 \\ (P_i - P_r)^{\frac{1}{2}}, u_{v_n} < 0 \end{cases}, \\ g_o(P_o, u_v) &= \begin{cases} (P_o - P_r)^{\frac{1}{2}}, u_{v_n} \ge 0 \\ (P_s - P_o)^{\frac{1}{2}}, u_{v_n} < 0 \end{cases}, \end{aligned} \tag{11}$$

where $P_s \in \mathbb{R}$ and $P_r \in \mathbb{R}$ represent the supply pressure and return pressure respectively.

3. Controller design

3.1. Backstepping control framework

Fig. 2 presents the control block diagram. To address the complex high-order dynamics of hydraulic manipulators, a backstepping control framework is adopted. The control design focuses on achieving asymptotic tracking performance. Firstly, a series of errors is defined as follows:

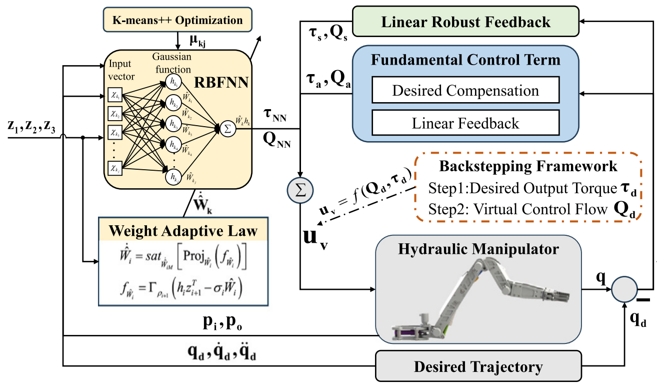

Fig. 2. Schematic diagram of overall control framework.

$$
\begin{aligned}
z_1 &= q(t) - q_d(t),\\
z_2 &= \dot{z}_1 + K_1 z_1,\\
z_r &= \dot{q}_d(t) - K_1 z_1,\\
z_3 &= F - F_d(t),
\end{aligned} \tag{12}
$$

where $z_1 \in \mathbb{R}^n$ represents the tracking error, $z_2 \in \mathbb{R}^n$ is the sliding mode variable, $K_1 \in \mathbb{R}^{n\times n}$ is the feedback gain matrix, $z_3 \in \mathbb{R}^n$ is the torque tracking error, and $F_d(t) \in \mathbb{R}^n$ is the desired output thrust, which will be designed later.

Assumption 1. $q_d(t)$ is a C^{n+2} desired trajectory such that $q_d(t), \dot{q}_d(t), \ddot{q}_d(t) \in L_\infty$.

Assumption 2. The modeling error $\tau_\varepsilon(t)$ and uncertain nonlinearities $d_{Q_o}(t), d_{Q_i}(t)$ can be divided into nominal values $\bullet^*$ and fast time-varying terms $\Delta\bullet$

$$
\begin{aligned}
\tau_\varepsilon(t) &= \tau_\varepsilon^*(t) + \Delta\tau_\varepsilon(t),\\
d_{Q_o}(t) &= d_{Q_o}^*(t) + \Delta d_{Q_o}(t),\\
d_{Q_i}(t) &= d_{Q_i}^*(t) + \Delta d_{Q_i}(t),
\end{aligned} \tag{13}
$$

Assumption 3. $\Delta\tau_\varepsilon(t)$, $\Delta d_{Q_o}(t)$, and $\Delta d_{Q_i}(t)$ are all bounded, i.e., $\|\Delta\tau_\varepsilon(t)\|_\infty \le \epsilon_1$, $\|\Delta d_{Q_o}(t)\|_\infty \le \epsilon_2$, $\|\Delta d_{Q_i}(t)\|_\infty \le \epsilon_3$, where $\epsilon_\bullet$, $\bullet \in \{1,2,3\}$ are unknown constants.

STEP I

Based on (1), (2), and (12), we can obtain

$$
\begin{aligned}
M\dot{z}_2 + Cz_2 &= \tau - \tau_f(\dot{q}) - M\dot{z}_r - Cz_r - G + \tau_\varepsilon(t)\\
&= \tau - \varXi_1^{\mathrm{T}} \odot \varTheta_1 - M\dot{z}_r - Cz_r - G + \Delta\tau_\varepsilon(t),
\end{aligned} \tag{14}
$$

where the linear regression matrix $\varXi_1$ and parameters matrix $\varTheta_1$ are defined as:

$$
\begin{aligned}
\varXi_1 &= [\dot{q},\ S_f(\dot{q}),\ -I_n]^{\mathrm{T}},\\
I_n &= [1, 1, \dots, 1]^{\mathrm{T}} \in \mathbb{R}^n,\\
\varTheta_1 &= [\varTheta_{1a},\ \varTheta_{1b},\ \varTheta_{1c}]^{\mathrm{T}} = [f_{\dot{q}},\ f_\mu,\ \tau_\varepsilon^*(t)]^{\mathrm{T}}.
\end{aligned} \tag{15}
$$

The use of measured signals poses a challenge due to sensor hardware limitations, which can only measure angles. To obtain angular velocity, differentiation is required, which amplifies noise and introduces time delay. To address this issue, ideal trajectory signals are utilized as a substitute for measured signals.

Theorem 1 (Lagrange's mean value theorem). *Let f be a real-valued function defined on the closed interval $[a,b]$, and suppose that f satisfies the following conditions:*

1. *f is continuous on the closed interval $[a,b]$,*
2. *f is differentiable on the open interval (a,b).*

Then, there exists at least one point $c \in (a,b)$ such that the instantaneous rate of change of the function at c equals the average rate of change over the interval $[a,b]$, i.e., $f'(c)(b-a) = f(b) - f(a)$

By applying the Mean Value Theorem, the following result can be obtained:

$$
S_f(\dot{q}) - S_f(\dot{q}_d) = g_s(\kappa_1, t) \odot \dot{z}_1, \quad \kappa_1 \in (\dot{q}, \dot{q}_d), \tag{16}
$$

where $g_s(\kappa_1, t) \in \mathbb{R}^n$ is a deterministic function. Considering the requirements for equation differentiability and continuity in controller design, we use $2atan(\gamma\bullet)/\pi$ as substitutes for the non-differentiable sign function $S_f(\bullet)$ in specific implementation. Thus, the specific expression of $g_s(\kappa_1, t)$ can be expressed as $g_s(\kappa_1, t) = \frac{2\gamma}{\pi\left(1+(\gamma\kappa_1)^2\right)}$, where γ is a manually set coefficient. Consequently, the desired output torque can be designed as follows:

$$
\begin{aligned}
\tau_d &= \tau_a + \tau_s + \tau_{NN},\\
\tau_a &= -K_{\alpha_1} z_2 + \varXi_{1d}^{\mathrm{T}} \odot \hat{\varTheta}_1 + M\dot{z}_r + Cz_r + G,
\end{aligned} \tag{17}
$$

where τ_a is the fundamental control thrust, the ideal signals matrix $\varXi_{1d} = [\dot{q}_d,\ S_f(\dot{q}_d),\ -I_n]^{\mathrm{T}}$, define $\tilde{\bullet} = \hat{\bullet} - \bullet$, where $\hat{\bullet}$ is the estimation of $\bullet$, $\tilde{\bullet}$ is the estimation error between $\bullet$ and $\hat{\bullet}$, the nonlinear robust feedback term τ_s and NN fitting term τ_{NN} will be designed in the next section. $K_{\alpha_1} \in \mathbb{R}^{n\times n}$ is the linear feedback gain matrix, which is sufficiently large such that N is positive definite:

$$
N = \begin{bmatrix} \rho_1 K_1^3 & -A \\ -A & 2\rho_2\left(K_{\alpha_1} + B - K_2\right) \end{bmatrix}, \tag{18}
$$

where

$$
\begin{aligned}
A &= \rho_1 K_1^2 + \rho_2 K_1 B,\\
B &= diag(\varTheta_{1b} \odot g_s(\kappa_1, t) + \rho_2 \varTheta_{1a}).
\end{aligned} \tag{19}
$$

where K_2 is an arbitrary positive diagonal matrix, $\rho_i \in \mathbb{R}, i \in \{1,2,3\}$ are the dimensional equilibrium coefficients.

By (14) and (17), we can obtain the first dynamic error equation,

$$
M\dot{z}_2 + Cz_2 = -K_{\alpha_1} z_2 + E + \varXi_{1d}^{\mathrm{T}} \odot \tilde{\varTheta}_1 + \Delta\tau_\varepsilon(t) + \tau_s + \tau_{NN}, \tag{20}
$$

where $E = (\varXi_{1d} - \varXi_1)^{\mathrm{T}} \odot \varTheta_1$.

Lemma 1. *For a block matrix*

$$
N = \begin{bmatrix} A & B \\ C & D \end{bmatrix}
$$

where $N \in \mathbb{R}^{(n+m)\times(n+m)}$, and A is invertible. The Schur complement of N with respect to A is defined as $NA^{-1} = D - CA^{-1}B$. Then, the matrix N is positive definite ($N > 0$) if and only if $A > 0$ and $NA^{-1} > 0$.

Remark 1. The positive definiteness of N is guaranteed by selecting a sufficiently large linear feedback gain $K_{\alpha 1}$. Based on the Schur Complement (presented in Lemma 1), $N > 0$ if $2\rho_2(K_{\alpha 1} + B - K_2) - A^T(\rho_1 K_1^3)^{-1} A > 0$. Since K_1, K_2, ρ_1, and ρ_2 are fixed design parameters, and A and B are bounded (due to Assumption 4), the right-hand side of the inequality $K_{\alpha 1} > \frac{1}{2\rho_2} A^T(\rho_1 K_1^3)^{-1} A - B + K_2$ is bounded. Thus, a sufficiently large choice of the design matrix $K_{\alpha 1}$ always exists to satisfy this condition.

STEP II

In the operating logic of a hydraulic manipulator, the flow rate is directly calculated as a voltage signal through static mapping (i.e., the virtual control flow rate Q_d is assumed to be equal to Q_L). This flow rate change leads to a corresponding change in thrust, which ultimately drives the motion of the manipulator. However, since the thrust designed in **STEP I** cannot be directly controlled, a backstepping method is introduced to design the flow rate accordingly.

By differentiating (12), we can obtain $\dot{z}_3 = \dot{F} - \dot{F}_d$. Considering (3), (9), and (13),

$$\beta_e^{-1}\dot{z}_3 = G_Q + Q_L - \beta_e \dot{F}_d - Q_{dn}^*(t) + \Delta d_Q(t), \tag{21}$$

where

$$\begin{aligned} G_Q &= -G_{\alpha_1} \odot \dot{x}_c + G_{\alpha_2}, \\ G_{\alpha_1} &= A_i \odot R_\alpha + A_o \odot R_\beta, \\ R_\alpha &= A_i \oslash V_i, \\ R_\beta &= A_o \oslash V_o, \\ G_{\alpha_2} &= \sigma(P_o - P_i) \odot (R_\alpha + R_\beta), \\ Q_L &= Q_i \odot R_\alpha + Q_o \odot R_\beta, \\ Q_{dn}^*(t) &= R_\alpha \odot d_{Q_i}^* + R_\beta \odot d_{Q_o}^*, \\ \Delta d_Q(t) &= R_\alpha \odot \Delta d_{Q_i} + R_\beta \odot \Delta d_{Q_o}. \end{aligned} \tag{22}$$

From (12), (14), and (17), it is easy to find that $\dot{F}_d$ is related to $q, \dot{q}, \ddot{q}, \Theta$ and t, so it can be further decomposed into an analyzable part $\dot{F}_{dc}$ and a non-analyzable part $\dot{F}_{du}$, as follows:

$$\begin{aligned} \dot{F}_d &= \dot{F}_{dc} + \dot{F}_{du}, \\ \dot{F}_{dc} &= \frac{\partial F_d}{\partial q}\dot{q} + \frac{\partial F_d}{\partial \dot{q}}\hat{\ddot{q}} + \frac{\partial F_d}{\partial t}, \\ \dot{F}_{du} &= \frac{\partial F_d}{\partial \dot{q}}(\ddot{q} - \hat{\ddot{q}}) + \frac{\partial F_d}{\partial \hat{\Theta}_1}\dot{\hat{\Theta}}_1. \end{aligned} \tag{23}$$

Therefore, Eq. (21) can be further written as

$$\beta_e^{-1}\dot{z}_3 = G_Q + Q_L - \Xi_2^{\mathrm{T}} \odot \Theta_2 - \beta_e^{-1}\dot{F}_{du} + \Delta d_Q(t), \tag{24}$$

where the linear regression matrix Ξ_2 and the parameter matrix Θ_2 are as follows:

$$\begin{aligned} \Xi_2 &= [\dot{F}_{dc},\ I_n]^{\mathrm{T}}, \\ \Theta_2 &= [\Theta_{2a},\ \Theta_{2b}]^{\mathrm{T}} = [\beta_e^{-1} I_n,\ Q_{dn}^*(t)]^{\mathrm{T}}. \end{aligned} \tag{25}$$

Thus, Q_d is designed to align with τ_d.

$$\begin{aligned} Q_d &= Q_a + Q_s + Q_b + Q_{NN}, \\ Q_s &= -G_Q - K_{\beta_1} z_3 + \Xi_2^{\mathrm{T}} \odot \hat{\Theta}_2, \\ Q_b &= -\frac{\rho_2}{\rho_3} z_2, \end{aligned} \tag{26}$$

where Q_a is the fundamental control flow, $K_{\beta_1} \in \mathbb{R}^{n\times n}$ is the linear feedback gain matrix, Q_b is the backstepping compensation term. So (24) can be rewritten as

$$\beta_e^{-1}\dot{z}_3 = -K_{\beta_1} z_3 + \Xi_2^{\mathrm{T}} \odot \tilde{\Theta}_2 - \beta_e^{-1}\dot{F}_{du} + \Delta d_Q(t) + Q_s + Q_b + Q_{NN}, \tag{27}$$

Thus, the second dynamic error equation is obtained.

3.2. NN approximation optimized by k-means++

The RBFNN algorithm is commonly used to estimate unknown uncertainties. Due to different object characteristics, this paper will estimate different objects, where the approximate function can be expressed as

$$\begin{aligned} F(\chi) &= [F_1(\chi_1), F_2(\chi_2), \dots, F_k(\chi_k)],\ k = 1, 2, \dots, P, \\ F_k(\chi_k) &= W_k^{\mathrm{T}} h_k(\chi_k) + \xi_k, \end{aligned} \tag{28}$$

where $\chi_k = [\chi_{k_1}, \chi_{k_2}, \dots, \chi_{k_j}] \in \mathbb{R}^j$ is the input vector for the k-th class, $W_k = [W_{k_1}, W_{k_2}, \dots, W_{k_j}] \in \mathbb{R}^{j\times n}$ is the weight vector for a network with j nodes, and $\xi_k \in \mathbb{R}^n$ is the residual term of the RBFNN, which satisfies $\|\xi_k\|_\infty < \delta_k$ and δ_k is an unknown bounded nonnegative constant.

The activation function of adaptive RBFNNs is the Gaussian function $h_k(\chi_k)$, which takes the following form:

$$\begin{aligned} h_k(\chi_k) &= [h_{k_1}(\chi_k), h_{k_2}(\chi_k), \dots, h_{k_j}(\chi_k)] \in \mathbb{R}^j, \\ h_{k_j}(\chi_k) &= \exp\left(-\frac{\|\chi_k - \mu_{k_j}\|_2^2}{2b_k^2}\right), \end{aligned} \tag{29}$$

where $b_k \in \mathbb{R}^P$ is the width of the Gaussian function, $\mu_{k_j} \in \mathbb{R}^j$ is the position of the hidden node, which will be optimized using the K-means++ algorithm. The entire expression represents an RBFNN function for a k-th class, n-DoF manipulator, with j nodes in the RBFNN.

Property 3. For the Gaussian activation function $h_{k_j}(\chi)$, it holds that $\|h_{k_j}(\chi)\|_\infty \le 1,\ \forall k, j$.

Remark 2. The selection of K-means++ over methods like Gaussian Mixture Model (GMM) or hierarchical clustering for RBFNN center optimization stems from practical trade-offs. While GMM may model more flexible cluster shapes, K-means++ offers significantly higher computational efficiency and simpler implementation, which is critical in control engineering applications. Furthermore, its objective of minimizing squared Euclidean distance aligns directly with the RBFNN's Gaussian basis function (29), since it is also distance-based, making it a suitable and effective method for this specific task.

Considering (20), (27), (28) and (29), the parameter uncertainty residual term and the modeling residual term can be modeled using RBFNNs,

$$\begin{aligned} \Xi_{1d}^{\mathrm{T}} \odot \tilde{\Theta}_1 + \Delta\tau_\epsilon(t) &= W_1^{\mathrm{T}} h_1(\chi_1) + \xi_1, \\ \Xi_2^{\mathrm{T}} \odot \tilde{\Theta}_2 - \beta_e^{-1}\dot{F}_{du} + \Delta d_Q(t) &= W_2^{\mathrm{T}} h_2(\chi_2) + \xi_2, \end{aligned} \tag{30}$$

The NN fitting term can be designed as follows:

$$\begin{aligned} \tau_{NN} &= -\hat{W}_1^{\mathrm{T}} h_1(\chi_1), \\ Q_{NN} &= -\hat{W}_2^{\mathrm{T}} h_2(\chi_2), \end{aligned} \tag{31}$$

Considering that the dynamic characteristics of τ_{NN} and Q_{NN} are different, and focusing on taking into account the system state variables, the input vectors χ_1 and χ_2 are defined as:

$$\begin{aligned} \chi_1 &= [z_1, z_2, q_d, \dot{q}_d, \ddot{q}_d], \\ \chi_2 &= [z_2, z_3, P_i, P_o], \end{aligned} \tag{32}$$

By substituting (31) into (20) and (27) we can obtain

$$M\dot{z}_2 + Cz_2 = -K_{\alpha_1} z_2 + E - \tilde{W}_1^{\mathrm{T}} h_1(\chi_1) + \xi_1 + \tau_s, \tag{33}$$

$$\beta_e^{-1}\dot{z}_3 = -K_{\beta_1} z_3 - \tilde{W}_2^{\mathrm{T}} h_2(\chi) + \xi_2 + Q_s. \tag{34}$$

Remark 3. The RBFNNs employed in (31) take $\chi_1 \in \mathbb{R}^{5n}$ and $\chi_2 \in \mathbb{R}^{4n}$ as inputs. According to universal approximation theory, RBFNNs provide a local approximation capability, which is only guaranteed over

Algorithm 1: RBFNN Algorithm with K-means++ Initialization.

Input: Offline data $\{\chi_k\}_{k=1}^{P}$, number of clusters j
Output: Cluster centers $\{\mu_k\}_{k=1}^{P}$, widths $\{b_k\}_{k=1}^{P}$

Step 1: Offline Data Collection
Use an unoptimized controller to collect data $\{\chi_k\}_{k=1}^{P}$.
Step 2: Normalize Data
Apply z-score normalization to the collected data $\{\chi_k\}_{k=1}^{P}$ to ensure all features (e.g., z_2, z_3, P_i, P_o) have zero mean and unit variance. This is critical for meaningful distance-based clustering.
Step 3: K-means++ Clustering
Randomly select the first cluster center μ_{k1} from the normalized data.
for $j = 2$ *to* j **do**
  Select μ_{kj} with probability $\propto D(\chi_k)^2$, where $D(\chi_k)$ is distance to nearest center.
  Assign normalized χ_k to nearest μ_{kj}; update μ_{kj} as mean of assigned points.
repeat
  Recompute cluster centers.
until μ_{kj} *converges*;
Step 4: Compute Width Parameters
Compute $b_k = \text{avg\_dist}/\sqrt{2}$ or $\text{max\_dist}/\sqrt{2}$ based on distances between final centers.

a predefined compact set $\mathcal{X}_c$. In our framework, this condition is formally guaranteed; the Lyapunov stability analysis in Theorem 2 proves that all error signals (z_1, z_2, z_3) are Semi-Globally Uniformly Ultimately Bounded (SGUUB), given the a priori bounds from Assumption 1 and Assumption 4, and physical hardware limits. Thus, the RBFNN inputs are guaranteed to remain within a convex and compact set $K = \{\chi \mid \|\chi\|_\infty \leq \chi_{max}\}$, satisfying the validity requirement for the approximation.

3.3. Adaptive law design

In order to enhance the robustness of the RBFNN controller, a dynamic weight vector estimation strategy was developed by introducing a modification term, as shown below:

$$\begin{aligned} \dot{\hat{W}}_i &= sat_{\dot{\hat{W}}_{iM}}\left[\mathrm{Proj}_{\hat{W}_i}\left(f_{\hat{W}_i}\right)\right], \\ f_{\hat{W}_i} &= \Gamma_{\rho_{i+1}}\left(h_i z_{i+1}^{\mathrm{T}} - \sigma_i \hat{W}_i\right), \ i = 1, 2, \end{aligned} \tag{35}$$

where $\Gamma_{\rho_{i+1}} \in \mathbb{R}^{j\times j}$ is the update gain matrix, it can be further written as $\Gamma_{\rho_{i+1}} = \Gamma_i \rho_{i+1}$, $\sigma_i \in \mathbb{R}^+$ denotes the modification coefficient, $sat_{\dot{\hat{W}}_{iM}}$ is the saturation function that limits the rate of $\hat{W}_i$ change, which can be expressed as

$$sat_{\dot{\hat{W}}_{iM}}(\varsigma) = \kappa\varsigma, \ \kappa = \begin{cases} 1, & \|\varsigma\| \leq \dot{\hat{W}}_{iM} \\ \frac{\dot{\hat{W}}_{iM}}{\|\varsigma\|}, & \|\varsigma\| > \dot{\hat{W}}_{iM}, \ i = 1, 2 \end{cases} \tag{36}$$

where $\dot{\hat{W}}_{iM}$ is a preset maximum update rate, $\mathrm{Proj}_{\hat{W}_i}(\bullet)$ is the discontinuous projection function, which can be expressed as

$$\mathrm{Proj}_{\hat{W}_i}(\bullet) = \begin{cases} 0, & \hat{W}_i = \overline{W}_i \text{ and } \bullet > 0, \\ 0, & \hat{W}_i = \underline{W}_i \text{ and } \bullet < 0, \ i = 1, 2, \\ \bullet, & \text{otherwise.} \end{cases} \tag{37}$$

where $\overline{W}_i$ and $\underline{W}_i$ represent the preset upper and lower bounds of $\hat{W}_i$,

Remark 4. The modifications in (35)–(37) are introduced to ensure adaptation robustness. The σ-modification prevents the parameter estimates $\hat{W}_i$ from drifting to infinity under disturbances or insufficient excitation. Meanwhile, the $\mathrm{Proj}(\cdot)$ and $sat(\cdot)$ operators enforce the prior knowledge of weight boundedness (Assumption 4). These mechanisms are essential for guaranteeing the boundedness of the estimation error $\tilde{W}_i$, which is a step in the stability proof of Theorem 2.

Property 4. The weight estimates $\hat{W}_i$ can be guaranteed to be within the preset bounded set $\overline{\Omega}_{W_i}$, i.e.,

$$\hat{W}_i \in \overline{\Omega}_{\hat{W}_i} = \{W_i : \underline{W}_i \leq \hat{W}_i \leq \overline{W}_i\}, \ i = 1, 2, \ \forall t. \tag{38}$$

Property 5. The weight update rate $\dot{\hat{W}}_i$ is bounded by

$$\|\dot{\hat{W}}_i\| \leq \dot{\hat{W}}_{iM}, \ i = 1, 2, \ \forall t. \tag{39}$$

Property 6. By considering the weight estimation error definition $\tilde{W}_i = \hat{W}_i - W_i$, and applying the fundamental properties of the Frobenius inner product and the matrix trace—specifically the norm definition $\|A\|_F^2 = \mathrm{tr}(A^T A)$ and the cyclic property of the trace $\mathrm{tr}(A^T B) = \mathrm{tr}(B^T A)$ —the following relationship is derived:

$$\begin{aligned} 2\mathrm{tr}(\tilde{W}_i^T \hat{W}_i) &= \mathrm{tr}\left(\tilde{W}_i^T\left(\tilde{W}_i + W_i\right)\right) + \mathrm{tr}\left(\left(\hat{W}_i - W_i\right)^T \hat{W}_i\right) \\ &= \mathrm{tr}\left(\tilde{W}_i^T \tilde{W}_i\right) + \mathrm{tr}\left(\hat{W}_i^T \hat{W}_i\right) + \mathrm{tr}\left(W_i^T\left(\tilde{W}_i - \hat{W}_i\right)\right) \\ &= \|\tilde{W}_i\|_F^2 + \|\hat{W}_i\|_F^2 - \|W_i\|_F^2 \end{aligned} \tag{40}$$

where $\|\cdot\|_F^2$ represents the Frobenius norm of a matrix and $\mathrm{tr}(\cdot)$ represents the trace of a matrix.

Assumption 4. The parameter matrices $\Theta_i, i \in \{1, 2\}$ in this paper have corresponding physical meanings, and considering the performance of the actual system, it is not difficult to conclude that the parameters are bounded, i,e.,

$$\Theta_i \in \overline{\Omega}_{\Theta_i} = \{\Theta_i : \Theta_{\mathrm{imin}} \leq \Theta_i \leq \Theta_{\mathrm{imax}}\}, \ i = 1, 2, \ \forall t. \tag{41}$$

Moreover, for the RBFNN, since its target is the actual physical system, the weights W can also be considered bounded,i,e,

$$W_i \in \overline{\Omega}_{W_i} = \{W_i : W_{\mathrm{imin}} \leq W_i \leq W_{\mathrm{imax}}\}, \ i = 1, 2, \ \forall t, \tag{42}$$

where Θ_{imin} and Θ_{imax} represent the lower and upper bounds of Θ_i, while W_{imin} and W_{imax} represent the lower and upper bounds of W_i.

3.4. Robust control term design

Most challenges were addressed through the design of backstepping and RBFNN. However, it is clear that there are residual terms present, as shown by Eqs. (33) and (34). Consequently, the linear robust feedback terms τ_s and Q_s must satisfy the following conditions:
Condition 1

$$\begin{aligned} & z_2^{\mathrm{T}} \tau_s \leq 0, \\ & z_2^{\mathrm{T}}(\xi_1 + \tau_s) < \eta_1. \end{aligned} \tag{43}$$

Condition 2

$$\begin{aligned} & z_3^{\mathrm{T}} Q_s \leq 0, \\ & z_3^{\mathrm{T}}(\xi_2 + Q_s) < \eta_2. \end{aligned} \tag{44}$$

where $\eta_1 \in \mathbb{R}^+$ and $\eta_2 \in \mathbb{R}^+$ are constants that affect the final convergence domain. By Young's inequality, we can obtain $2z_\bullet^{\mathrm{T}}\xi_\bullet \leq z_\bullet^{\mathrm{T}} z_\bullet + \xi_\bullet^{\mathrm{T}}\xi_\bullet$. Therefore, the value of $\eta_\bullet$ can be determined by $\eta_\bullet = \left(\|z_\bullet\|^2 + \delta_\bullet^2\right)/2$.

This is combined with (33), (34), and *Property 4*. The τ_s and Q_s can be designed as

$$\tau_s = - K_{\alpha_2} z_2,\ K_{\alpha_2} = \frac{(\delta_1 + \nu_1)^2}{4\eta_1},$$
$$Q_s = - K_{\beta_2} z_3,\ K_{\beta_2} = \frac{(\delta_2 + \nu_2)^2}{4\eta_2}, \tag{45}$$

where $\nu_1 \in \mathbb{R}$ and $\nu_2 \in \mathbb{R}$ are arbitrarily small positive constants.

Theorem 2. *Considering the error dynamics (33) (34), robustness conditions (43) (44), adaptive laws (35), and Frobenius inner product property (40). The positive-definite-function V_s is defined as*

$$V_s = \frac{1}{2} z_1^T \rho_1 K_1^2 z_1 + \frac{1}{2} z_2^T \rho_2 M z_2 + \frac{1}{2} \beta_e^{-1} z_3^T \rho_3 z_3 + \frac{1}{2} \sum_{k=1}^{2} tr\left(\tilde{W}_k^T \Gamma_k^{-1} \tilde{W}_k\right), \tag{46}$$

Since the system satisfies the uniformly ultimately bounded (UUB) condition, the Lyapunov function $V_s(t)$ is bounded by:

$$V_s(t) \le e^{-\mu^* t} V_s(0) + \frac{\xi}{\mu^*}(1 - e^{-\mu^* t}),\ \forall t \ge 0. \tag{47}$$

This implies that the tracking errors converge exponentially to a compact set Ω_r, defined as:

$$\Omega_r = \left\{ \mathbf{z} \;\middle|\; \lim_{t \to \infty} V_s(\mathbf{z}, t) \le \frac{\xi}{\mu^*} \right\}, \tag{48}$$

where

$$\mu^* = \min \left\{ \begin{array}{l} \lambda_{\min}(K_1),\ \dfrac{2\lambda_{\min}(K_2)}{\lambda_{\max}(M)}, \\ 2\beta_e \lambda_{\min}(K_{\beta_1}),\ \dfrac{\rho_{i+1}\sigma_i}{\lambda_{\max}(\Gamma_i^{-1})}, i = 1, 2 \end{array} \right\},$$
$$\xi = \rho_2 \eta_1 + \rho_3 \eta_2 + \frac{1}{2} \sum_{k=1}^{2} \rho_{k+1} \sigma_k \|W_k\|_F^2. \tag{49}$$

Furthermore, the ultimate convergence threshold size depends on the term ξ/μ^* (i.e., $V_s(\infty) \le \xi/\mu^*$). Therefore, by adjusting the control gain and the parameter update law, the size of the residual set Ω_r can be arbitrarily reduced as $t \to \infty$. □

Proof of Theorem 2 Define the Lyapunov function V_s as

$$V_s = \frac{1}{2} z_1^T \rho_1 K_1^2 z_1 + \frac{1}{2} z_2^T \rho_2 M z_2 + \frac{1}{2} \beta_e^{-1} z_3^T \rho_3 z_3 + \frac{1}{2} \sum_{k=1}^{2} tr\left(\tilde{W}_k^T \Gamma_k^{-1} \tilde{W}_k\right). \tag{50}$$

The derivative of V_s can be calculated as:

$$\begin{aligned}
\dot{V}_s &= z_1^T \rho_1 K_1^2 \dot{z}_1 + z_2^T \rho_2 \left(M \dot{z}_2 + \dot{M} z_2\right) + \beta_e^{-1} z_3^T \rho_3 \dot{z}_3 + \sum_{k=1}^{2} tr\left(\tilde{W}_k^T \Gamma_k^{-1} \dot{\hat{W}}_k\right) \\
&\le z_1^T \rho_1 K_1^2 \dot{z}_1 + z_2^T \rho_2(-K_{\alpha_1} z_2 + E) - z_3^T \rho_3 K_{\beta_1} z_3 \\
&\quad + z_2^T \rho_2(-\tilde{W}_1^T h_1(\chi_1) + \xi_1 + \tau_s) + z_3^T \rho_3(-\tilde{W}_2^T h_2(\chi_2) \\
&\quad + \xi_2 + Q_s) + \sum_{k=1}^{2} tr(\tilde{W}_k^T \Gamma_k^{-1} \mathrm{Proj}_{\hat{W}_k}(\Gamma_{\rho_{k+1}}(h_k z_{k+1}^T - \sigma_k \hat{W}_k))) \\
&\le -\frac{1}{2} z_1^T \rho_1 K_1^3 z_1 - z_2^T \rho_2 K_2 z_2 - \frac{1}{2} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} N \begin{bmatrix} z_1 & z_2 \end{bmatrix} \\
&\quad - z_3^T \rho_3 K_{\beta_1} z_3 + z_2^T \rho_2(\xi_1 + \tau_s) + z_3^T \rho_3(\xi_2 + Q_s) - \sum_{k=1}^{2} \rho_{k+1} \sigma_k tr\left(\tilde{W}_k^T \hat{W}_k\right) \\
&\le -\frac{1}{2} z_1^T \rho_1 K_1^3 z_1 - z_2^T \rho_2 K_2 z_2 - z_3^T \rho_3 K_{\beta_1} z_3 + \rho_2 \eta_1 \\
&\quad + \rho_3 \eta_2 - \frac{1}{2} \sum_{k=1}^{2} \rho_{k+1} \sigma_k \|\tilde{W}_k\|_F^2 + \frac{1}{2} \sum_{k=1}^{2} \rho_{k+1} \sigma_k \|W_k\|_F^2 \\
&\le -\mu^* V_s + \xi.
\end{aligned} \tag{51}$$

By solving the inequality, we can derive the compact set Ω_r in (48), which indicates that all signals are bounded and capable of achieving exponential tracking performance. *Theorem 2* is thereby proven, showing that the designed control law ensures closed-loop stability even in the presence of model nonlinearities and parameter uncertainties. □

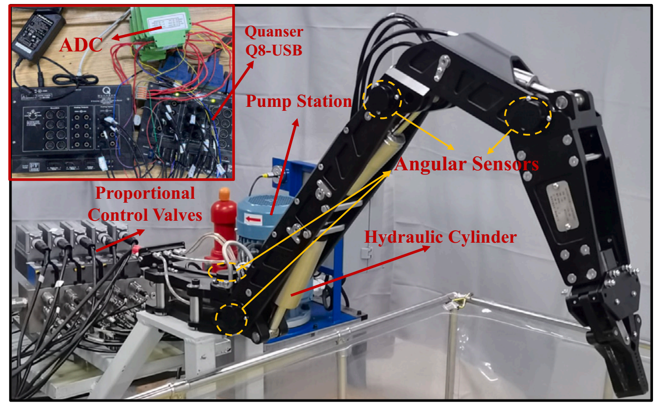

Fig. 3. Hydraulic manipulator hardware platform.

4. Experiment analysis

4.1. Hardware platform

To validate the performance of the proposed control strategy, a 4-DoF hydraulic manipulator was employed in the experiment (as shown in Fig. 3). The manipulator consists of a swing joint, boom joint, stick joint, wrist joint, and a gripper at the end-effector. The linear motion of hydraulic actuators drives each joint's rotation, and proportional valves regulate the output flow rate (4WRPEH6 C3B12L-3X/M/24A1). Joint angles are measured using high-accuracy angle sensors (Melexis MLX90316), which operate in PWM output mode. Pressure data are collected using high-precision pressure sensors (GEFRAN KS-E-E-Z-B16D-M-V-530), offering a resolution of 1.5 Pa and a response time of less than 1 ms. A total of 10 sensors (distributed in 5 groups) are employed to measure the pressure in the head-end and rod-end chambers of each hydraulic cylinder. The hydraulic system features a pump-driving motor (ABB M2BAX-112MA-4) with a rated power of 4 kW and a maximum output flow rate of 12 L/min. Quarc QPID-e and Q8-USB are utilized to process the data collected by the sensors. MATLAB/SIMULINK is used as the PC/Host Computer. The Host PC is equipped with a 13th Gen Intel(R) Core(TM) i7-13700 processor and 32.0 GB of RAM. The overall program's sampling frequency is 1000 Hz.

4.2. Experiment setup

To demonstrate the effectiveness of the proposed control strategy, validation was conducted on the wrist and stick joints, while a trajectory (as shown in Fig. 4) containing various dynamic characteristics was designed (static, constant angular velocity, constant angular acceleration, and constant angular jerk) to verify the controller's overall asymptotic tracking performance.

To demonstrate the efficacy of the proposed control strategy, the following four controllers are selected for comparative analysis (the gains of the controllers were adjusted to the critical stability limit until slight jittering occurred):

- **C-4**: This is the proposed controller in this paper. The control law is represented by (12), (17), (26), (35) and (31), the parameters were selected as $K_1 = diag[80, 120]$, $K_{\alpha_1} = diag[140, 50]$, $K_{\beta_1} = diag[140, 50]$, $\Gamma_{\rho_2} = diag[500, 500]$, $\Gamma_{\rho_3} = diag[0.003, 0.005]$, $\sigma_1 = 0.5$, $\sigma_2 = 1$, $\hat{W}_1(0) = \hat{W}_2(0) = 0$, $b_1 = 8 \times 10^5$, $b_2 = 1 \times 10^6$, and the number of nodes in the single hidden layer of the RBFNN was set to 5.
- **C-3**: This is a controller with adaptive model compensation, where the parameter settings are consistent with C-4. Compared with C-4, in dealing with uncertain terms, the parameter estimation values ($\hat{\Theta}_i$) are updated through a gradient- descent-based adaptive law [43], i.e.,

$$\dot{\hat{\Theta}}_i = \text{Proj}\left[\gamma_i \Xi_i diag(z_{i+1})\right], \tag{52}$$

where $\gamma_1 = diag[150, 50, 50]$ and $\gamma_2 = diag[5 \times 10^{-7}, 1 \times 10^{-7}]$ are the adaptive gains.
- **C-2**: This is a controller with deterministic model compensation. Compared to C-3, this controller maintains other parameters unchanged but does not update the estimated parameters, i.e., $\dot{\hat{\Theta}}_i \equiv 0$.
- **C-1**: This is a sliding mode controller. Compared with C-2, the estimated parameters are equal to 0, i.e., $\hat{\Theta}_i \equiv 0$.

In terms of control effectiveness, the following indicators are used for comparison: maximum tracking error ($e_M = \max\{e\}$), root-mean-square-error (RMSE: $\sqrt{\frac{1}{T}\int_0^T |e|^2 dt}$), normalized performance indicator ($\rho_e = e_M / \|\dot{q}_d\|_\infty$), and integral-absolute-error (IAE: $\int_0^T |e| dt$).

4.3. Experiment results

Figs. 4 and 5 depict the tracking trajectories and error profiles of the stick and wrist joints, respectively. As observed, the non-model-based controller C-1 exhibits the most significant tracking errors. Although its feedback gains were tuned to their stability limits, the absence of dynamic compensation prevents it from effectively suppressing nonlinearities, and thus preventing further gain increases without inducing oscillations.

In contrast, C-2 achieves a substantial improvement in tracking accuracy by incorporating a model-based feedforward compensation term (i.e., $\Xi_{\bullet} \odot \hat{\Theta}_{\bullet}$). This term effectively mitigates errors during dynamic transitions. However, since the physical parameters cannot be identified with perfect precision offline (i.e., $\hat{\Theta}_{\bullet} \neq \Theta_{\bullet}$), C-2 suffers from residual steady-state errors caused by parametric mismatches. C-3 addresses this limitation by employing a gradient-descent-based adaptive law (52) to estimate uncertain parameters online. While this strategy further reduces tracking errors compared to C-2, its performance is constrained by the adaptation mechanism itself: since the update law is driven solely by tracking errors, it is susceptible to parameter drift or sluggish adaptation, particularly in the presence of sensor noise or small error signals.

While the overall error metrics summarized in Table 1 demonstrate the statistical superiority of the proposed method, the zoomed-in views in Figs. 4 and 5 provide crucial insights into the controller performance under critical dynamic conditions.

Specifically, the enlarged sub-plot in Fig. 4 highlights the tracking performance of the stick joint during the motion reversal phase (around $t = 18.3$ s). At this zero-velocity crossing point, hydraulic manipulators typically suffer from severe nonlinearities caused by the valve dead-zone and Stribeck friction. It can be observed that the error curve of C-3 exhibits noticeable fluctuations (chattering) during this transition, indicating that the traditional gradient-based adaptation law struggles to compensate for fast-varying friction dynamics instantaneously. In contrast, the error profile of the proposed C-4 remains smooth and bounded near zero. This confirms that the optimized RBFNN effectively captures the localized inverse dynamics, significantly mitigating the impact of "stick-slip" phenomena during direction changes.

Similarly, the zoomed-in region in Fig. 5 (ranging from 13.5 s to 15.5 s) details the transient response of the wrist joint during a rapid trajectory tracking phase. C-1 and C-2 show evident steady-state errors and phase lags due to the lack of effective compensation for unmodeled uncertainties. Although C-3 reduces the steady-state error, it exhibits slower convergence when settling into the desired trajectory. Conversely, C-4 demonstrates superior transient performance with faster convergence and negligible overshoot. This improved adaptability is

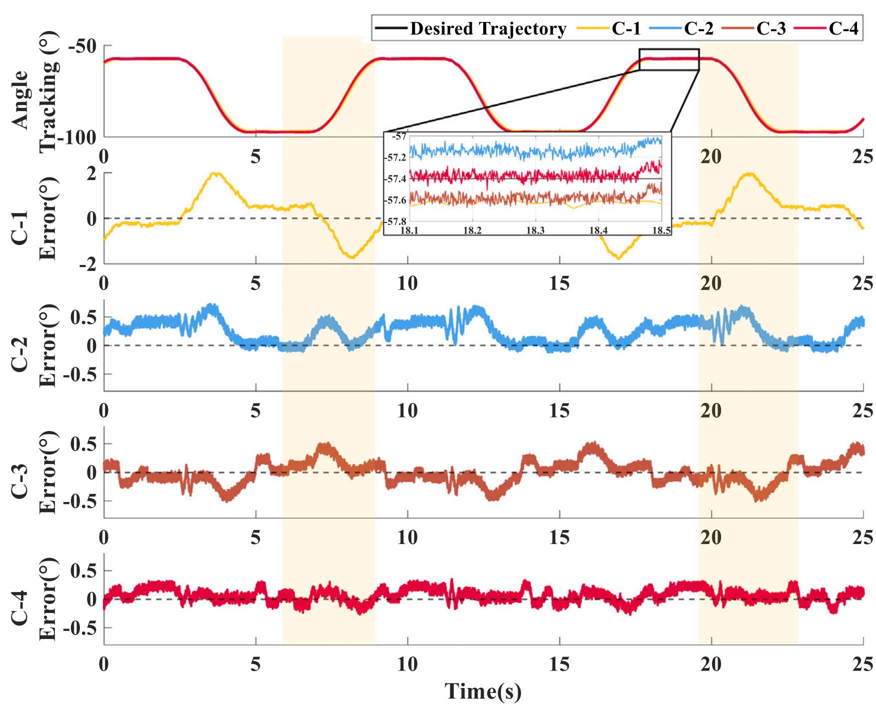

Fig. 4. The tracking performance of stick joint.

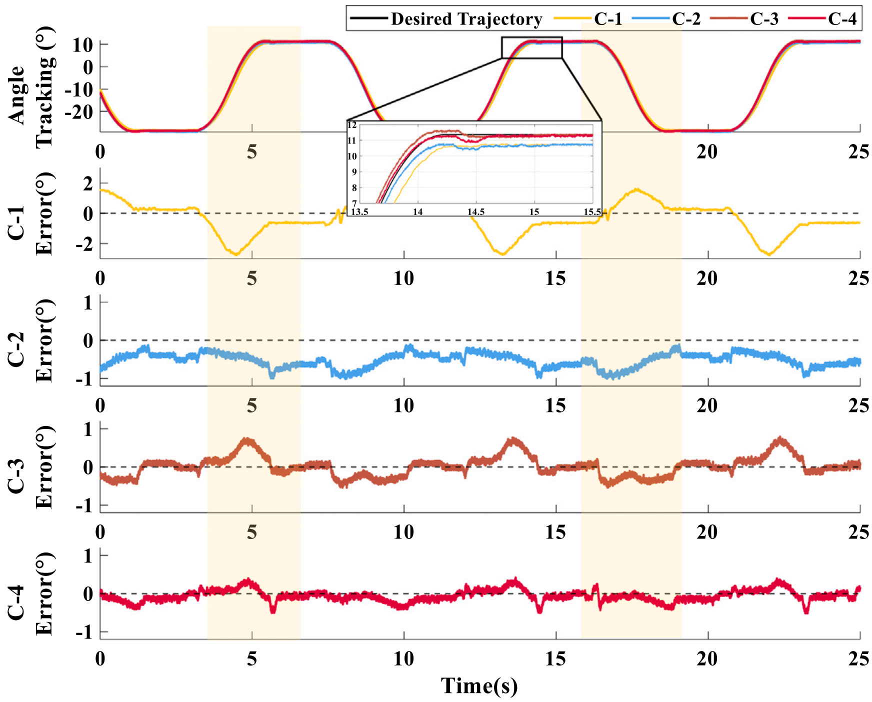

Fig. 5. The tracking performance of wrist joint.

Table 1
Numerical comparison of tracking results for different controllers.

		e_M (deg)	RMSE (deg)	ρ_e (s)	IAE (deg·s)
Stick Joint	C-1	2.0884	0.8942	3.9886	18,809.6344
	C-2	0.7275	0.3084	1.3893	6619.0273
	C-3	0.5302	0.1967	1.0126	4145.5564
	C-4	0.3510	0.1246	0.6703	2663.7473
Wrist Joint	C-1	2.7820	1.0821	5.3132	22,353.4146
	C-2	1.0435	0.5763	1.9929	14,315.5313
	C-3	0.7942	0.2694	1.5168	5474.2577
	C-4	0.5190	0.1601	0.9912	3405.9601

attributed to the K-means++ algorithm, which optimizes the distribution of neuron centers based on the system's operational space, thereby ensuring higher approximation accuracy for unmodeled residuals.

This enhanced performance is quantitatively validated by Table 1. The RMSE metrics, calculated over the entire multi-cycle trajectory, show a substantial improvement. Compared to the C-3 (ARC) controller, the proposed C-4 method achieves a 36.7% reduction in RMSE for the Stick Joint (from 0.1967 deg to 0.1246 deg) and a 40.6% reduction in RMSE for the Wrist Joint (from 0.2694 deg to 0.1601 deg). This significant improvement in tracking precision, both in average metrics and during critical dynamic transitions, underscores the practical superiority of the optimized NN-assisted approach. Ultimately, C-4 achieves the most precise tracking performance and the best error suppression among all controllers. The tracking results demonstrate that C-4 is well-suited for scenarios demanding precise control and adaptability.

Figs. 6 and 7 illustrate the control outputs of the stick joint and wrist joint under C-3 and C-4, respectively. From top to bottom, the figures display the desired output torque with its NN compensation term ($\tau_{Ld}/\tau_{C-3/4}$), and the output voltage (u_v). It can be seen that the outputs of C-4 exhibit smaller fluctuations and stronger stability compared to those of C-3. In particular, the C-3 method demonstrates larger fluctuation amplitude, especially in thrust control, where its performance is notably inferior. This highlights the limitations of C-3 in handling rapid dynamic changes and uncertainty compensation effectively.

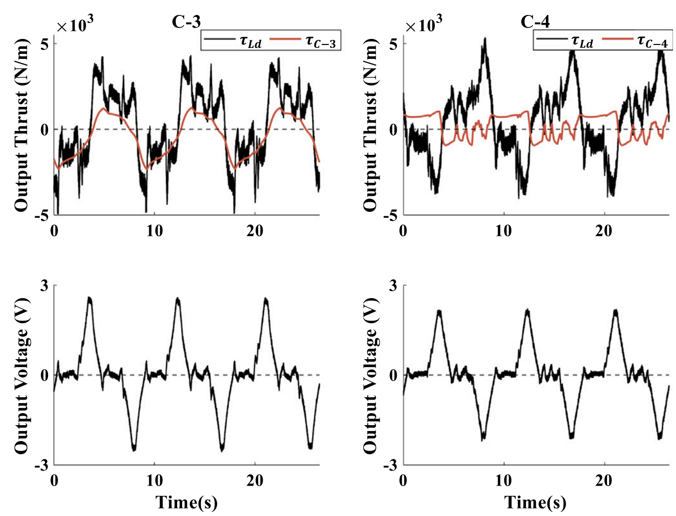

Fig. 6. Desired output torque and voltage output of stick joint.

In terms of output voltage, both methods exhibit comparable performance with regular temporal fluctuations. However, the output voltage of C-4 is slightly lower than that of C-3. More importantly, C-4 shows a superior capability in suppressing voltage transients near peaks and during abrupt voltage changes, such as those observed at 8 s and 16 s. This enhanced performance is further corroborated by C-4's lower tracking errors in dynamic transitions (as analyzed before). Overall, the C-4 method demonstrates significant improvements over C-3 in dynamic response, tracking accuracy, and robustness. These advantages establish C-4 as a more effective and reliable control strategy, particularly in applications requiring high precision and adaptability.

4.4. Conclusion

This paper presents a novel NN-assisted adaptive robust control framework designed to achieve high-precision motion control for

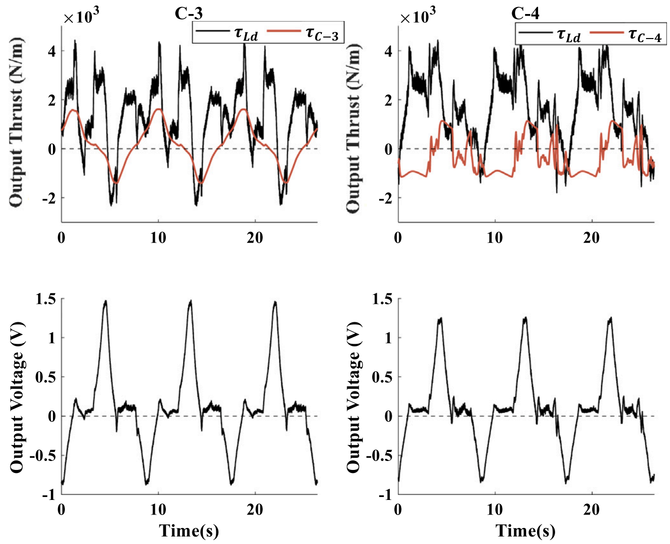

Fig. 7. Desired output torque and voltage output of wrist joint.

multi-DoF hydraulic manipulators operating under parametric uncertainties and significant unmodeled nonlinearities. A backstepping approach was utilized to systematically manage the system's high-order, coupled dynamics. The core contribution is a hybrid compensation strategy: an RBFNN, structurally optimized by the K-means++ algorithm, compensates for complex unmodeled dynamics, while a modification term and the desired compensation concept are integrated to ensure robustness against parameter drift and sensor noise. Rigorous theoretical analysis has been conducted to establish the asymptotic tracking performance of the closed-loop system. Comparative experimental results obtained on a multi-DoF hydraulic manipulator validated the proposed controller's effectiveness and demonstrated superior tracking precision over baseline control strategies.

While this study validates the proposed approach, its limitations indicate directions for future research. The controller was validated in an environment without significant unstructured external disturbances and did not explicitly address non-stationary operating conditions. Furthermore, the K-means++ optimization is performed offline, and the adaptation relies on a projection operator with pre-defined bounds. Future work will therefore focus on extending this strategy to these more complex and dynamic environments, as well as exploring advanced FATs such as the (p,q)-analogue of Bernstein-Stancu or Mastroianni operators, and integrating deep neural networks (DNNs) or reinforcement learning to enhance the controller's online adaptability.

CRediT authorship contribution statement

Manzhi Qi: Writing – original draft, Software, Methodology, Formal analysis. **Yangxiu Xia:** Software, Investigation. **Shizhao Zhou:** Writing – review & editing, Methodology. **Litong Lyu:** Writing – review & editing, Methodology, Investigation, Funding acquisition. **Zheng Chen:** Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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